Predictive Customer Churn Model for ConnectTel

Project Report

1. Project Overview

This project develops an end-to-end machine learning workflow to predict customer churn for ConnectTel, a telecom company. The goal is to identify customers who may be at higher risk of leaving and provide interpretable insights that can support proactive retention activities.

2. Business Problem & Objective

Customer churn can cause recurring revenue loss and increase the cost of acquiring replacement customers. The objective is to build a classification model that predicts whether a customer is likely to churn, while also understanding the factors associated with churn so that retention teams can take informed action.

3. Dataset & Data Preparation

The project uses the Telco Customer Churn dataset containing customer demographics, services, account information, contract details, charges, tenure and the churn target. The dataset contains 7,043 records and 21 original columns. TotalCharges was converted from text to numeric form and missing values were handled. CustomerID was removed before model training.

4. Exploratory Data Analysis

Contract: Month-to-month customers show 42.71% observed churn, compared with 11.27% for one-year and 2.83% for two-year contracts. This indicates a strong observed relationship between contract type and churn.

Internet service: Fiber optic customers show 41.89% observed churn compared with 18.96% for DSL customers. This is an observed association and does not establish that the service type causes churn.

Tenure and charges: Churned customers have a mean tenure of about 18 months compared with about 38 months for retained customers. Churned customers also have higher average monthly charges.

5. Feature Engineering

Three engineered features were created: **TotalChargesPerTenure**, which relates total charges to tenure; **ServiceCount**, which counts subscribed services; and **TenureGroup**, which groups customers by tenure range. Categorical variables were one-hot encoded and numerical variables were standardized within preprocessing pipelines.

6. Modeling

Three classification models were trained: Logistic Regression, Random Forest and XGBoost. The workflow used an 80/20 stratified train-test split, stratified 5-fold cross-validation and GridSearchCV hyperparameter tuning. ROC-AUC was used as the main optimization metric because the churn target is imbalanced.

7. Final Test Results

Model	Accuracy	Precision	Recall	F1	AUC-ROC
XGBoost	0.7999	0.6597	0.5080	0.5740	0.8472
Random Forest	0.7551	0.5266	0.7674	0.6246	0.8414
Logistic Regression	0.8041	0.6531	0.5588	0.6023	0.8413

XGBoost achieved the highest held-out test ROC-AUC in this experiment (0.8472) and was selected for SHAP interpretation. Random Forest produced the highest recall (0.7674), while Logistic Regression produced the highest accuracy (0.8041). These metrics show different trade-offs between detecting churners and limiting false positives.

8. Model Interpretation — SHAP

SHAP (SHapley Additive exPlanations) was used to interpret the selected XGBoost model. The SHAP analysis summarizes how input features contribute to model predictions and helps connect model behavior with business questions about churn drivers.

9. Business Insights & Recommendations

• Focus retention analysis on high-risk month-to-month customers and test suitable incentives for longer-term contracts.
• Strengthen onboarding and engagement during the early customer lifecycle.
• Review high-charge customers with elevated predicted churn risk for plan/value issues and suitable loyalty options.
• Investigate the higher observed churn among fiber optic customers through pricing, service quality, technical support and competitive factors.
• Test relevant Online Security and Tech Support bundles or support offers for appropriate customers.

10. Conclusion

The project provides a complete churn-prediction workflow from data preparation and EDA through feature engineering, model training, tuning, evaluation and SHAP interpretation. The resulting workflow connects customer data to churn risk identification and interpretable retention actions. The model should be treated as a decision-support tool and validated with real customer behavior and controlled retention experiments before operational deployment.

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, SHAP and Jupyter Notebook.